

JKBOSE 2026

CLASS 11TH

BINOMIAL THEOREM

MATHS

LECTURE-04

BY-MEHRAJ SIR



TOPICS TO BE COVERED

1

Miscellaneous Exercise

2

Related Problems

QUESTION (NCERT EXAMPLAR)

The two successive terms in the expansion of $(1+x)^{24}$ whose coefficients are in the ratio $1:4$ are

1 $\times$ 3rd and 4th

2 $\times$ 4th and 5th

3 $\checkmark$ 5th and 6th

4 $\times$ 6th and 7th

(Sol) Here $T_{r+1} = {}^{24}C_r x^r$
 $\& T_{r+2} = T_{r+1+1} = {}^{24}C_{r+1} x^{r+1}$

Here, It's given that;

$$\frac{{}^{24}C_r}{{}^{24}C_{r+1}} = \frac{1}{4}$$

$$\Rightarrow \frac{{}^{24}C_r}{{}^{24}C_{r+1}} = \frac{(24-r)! (r+1)!}{(24-r-1)! r!} = \frac{1}{4}$$

$$\Rightarrow \frac{(24-r)(24-r-1) \dots (r+2)(r+1)}{(24-r)(24-r-1) \dots (r+2)(r+1)} = \frac{1}{4}$$

$$\Rightarrow \frac{r+1}{24-r} = \frac{1}{4}$$

$$\Rightarrow 4r+4 = 24-r$$

$$\Rightarrow 5r = 20$$

$$\Rightarrow r = 4$$

So, Required Terms are T_5 & T_6

QUESTION (NCERT EXAMPLAR)

If the coefficients of 2nd, 3rd and the 4th terms in the expansion of $(1+x)^n$ are in A.P., then value of n is

1

2X

2

7✓

3

11X

4

14X

(Sol) Here $T_2 = {}^nC_1 x^1$

$T_3 = {}^nC_2 x^2$

$T_4 = {}^nC_3 x^3$

Since, the co-efficients of x, x^2 & x^3 are in A.P. i.e. ${}^nC_1, {}^nC_2$ & nC_3 are in A.P.

So, ${}^nC_2 - {}^nC_1 = {}^nC_3 - {}^nC_2$

$\Rightarrow 2{}^nC_2 = {}^nC_3 + {}^nC_1$

$$\Rightarrow \frac{2 \times n!}{(n-2)! 2!} = \frac{n!}{(n-3)! 3!} + \frac{n!}{(n-1)! 1!}$$

$$\Rightarrow \frac{2 n(n-1)(n-2)!}{(n-2)! \cdot 2} = \frac{n(n-1)(n-2)(n-3)!}{(n-3)! \cdot 6} + \frac{n(n-1)!}{(n-1)!}$$

$$\Rightarrow n(n-1) = \frac{n(n-1)(n-2)}{6} + n$$

$$\Rightarrow n(n-1) = \left(\frac{(n-1)(n-2)}{6} + 1 \right)$$

$$\Rightarrow 6n-6 = n^2-3n+2+6$$

$$\Rightarrow n^2-3n+8-6n+6=0$$

$$\Rightarrow n^2 - 9n + 14 = 0$$

$$\Rightarrow (n-7)(n-2) = 0.$$

$$\Rightarrow n = 7, 2.$$

QUESTION (NCERT EXAMPLAR)

If A and B are coefficient of x^n in the expansions of $(1+x)^{2n}$ and $(1+x)^{2n-1}$ respectively, then $\frac{A}{B}$ equals

- 1 x (Sol) Here $T_{r+1} = {}^{2n}C_r x^r$
- 2 $\checkmark$ & $T'_{r+1} = {}^{2n-1}C_r x^r$
So, $A = {}^{2n}C_n$ & $B = {}^{2n-1}C_n$
- 3 $1/2$ $\Rightarrow A = \frac{(2n)!}{(2n-n)!n!}$ & $B = \frac{(2n-1)!}{(2n-1-n)!n!}$
- 4 $1/n$

$$\begin{aligned} \Rightarrow \frac{A}{B} &= \frac{(2n)!}{n!n!} \times \frac{(n-1)!n!}{(2n-1)!} \\ &= \frac{2n(2n-1)!}{n(n-1)!} \times \frac{(n-1)!}{(2n-1)!} \\ &\Rightarrow \boxed{\frac{A}{B} = 2} \end{aligned}$$

QUESTION (JEE-MAINS 2024)

If the term independent of x in the expansion of $\left(\sqrt{ax^2} + \frac{1}{2x^3}\right)^{10}$ is **105**, then a^2 is equal to

1

(Sol) Here $T_{r+1} = {}^{10}C_r (\sqrt{a}x^2)^{10-r} \left(\frac{1}{2x^3}\right)^r$
 $\Rightarrow T_{r+1} = {}^{10}C_r \frac{(\sqrt{a})^{10-r}}{2^r} \times \frac{x^{20-2r}}{x^{3r}}$

2

$\Rightarrow T_{r+1} = {}^{10}C_r \frac{(\sqrt{a})^{10-r}}{2^r} \cdot x^{20-5r}$

3

For Term independent of x , we have;
 $20 - 5r = 0$

4

$\Rightarrow r = 4$

So, Constant term = ${}^{10}C_4 \frac{(\sqrt{a})^{10-4}}{2^4}$
 $\Rightarrow {}^{10}C_4 \frac{(a^{\frac{1}{2}})^6}{16} = 105$

$\Rightarrow \frac{10!}{6!4!} \frac{a^3}{16} = 105$

$\Rightarrow \frac{10 \times 9 \times 8 \times 7 \times 6!}{6! \times 24} a^3 = 105 \times 16$

$\Rightarrow 210 a^3 = 105 \times 16$

$a^3 = \frac{105 \times 16}{210}$
 $\Rightarrow a = 2$
 So, $a^2 = 4$

QUESTION (JEE-MAINS 2021)

The term independent of ' x ' in the expression of $\left(\frac{x+1}{x^{\frac{2}{3}} - x^{\frac{1}{3}} + 1} - \frac{x-1}{x - x^{\frac{1}{2}}} \right)^{10}$, where $x \neq 0, 1$ is equal to _____

QUESTION (JEE-MAINS 2021)

The sum of all those terms which are rational numbers in the expansion of $(2^{\frac{1}{3}} + 3^{\frac{1}{4}})^{12}$ is:

1 ~~X~~ 89

2 ~~X~~ 35

3 ~~X~~ 27

4 ✓ 43

(Sol) Here $T_{r+1} = {}^{12}C_r (2^{\frac{1}{3}})^{12-r} (3^{\frac{1}{4}})^r$.

$$\Rightarrow T_{r+1} = {}^{12}C_r (2)^{\frac{12-r}{3}} \cdot (3)^{\frac{r}{4}}$$

For rational terms, we have

$$r = 0, 12$$

$$\Rightarrow \text{Required Sum} = {}^{12}C_0 (2)^{\frac{12-0}{3}} \cdot (3)^{\frac{0}{4}} + {}^{12}C_{12} (2)^{\frac{12-12}{3}} (3)^{\frac{12}{4}}$$

$$\Rightarrow \text{Required Sum} = 16 + 27 = 43.$$

QUESTION (JEE-MAINS 2021)

If ${}^{20}C_r$ is the coefficient of x^r in the expansion of $(1+x)^{20}$, then the value of $\sum_{r=0}^{20} r^2 {}^{20}C_r$ is equal to

1 380×2^{18}

2 420×2^{18}

3 420×2^{19}

4 380×2^{19}

(Sol) From Binomial Theorem, we have;

$$(1+x)^{20} = \sum_{r=0}^{20} {}^{20}C_r x^r \quad \text{--- (1)}$$

Differentiating (1) w.r.t. x on b/s, we get;

$$20(1+x)^{19} = \sum_{r=0}^{20} {}^{20}C_r r x^{r-1}$$

$$\Rightarrow 20(1+x)^{19} x = \sum_{r=0}^{20} {}^{20}C_r r x^r \quad \text{--- (2)}$$

Differentiating (2) w.r.t. x on b/s, we get

$$20 \left\{ (1+x)^{19} \times 1 + x \times 19(1+x)^{18} \right\} = \sum_{r=0}^{20} {}^{20}C_r r^2 x^{r-1}$$

$$\Rightarrow \sum_{r=0}^{20} r^2 {}^{20}C_r x^{r-1} = 20 \left\{ (1+x)^{19} + 19x(1+x)^{18} \right\}$$

Put $x=1$, we get;

$$\sum_{r=0}^{20} r^2 {}^{20}C_r = 20 \left\{ 2^{19} + 19 \times 2^{18} \right\}$$

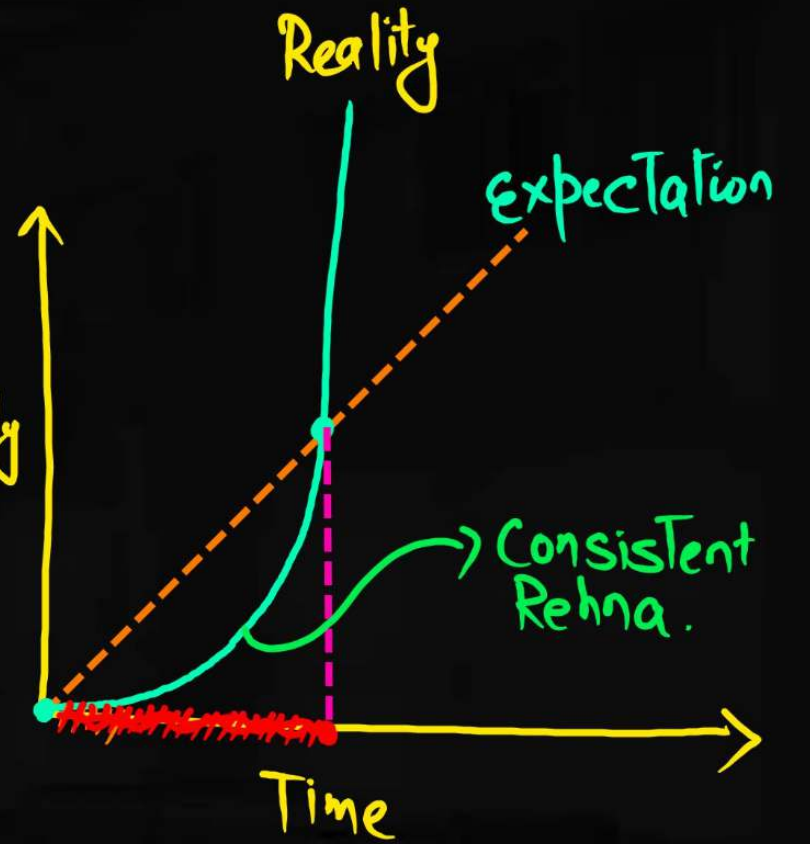
$$\Rightarrow \sum_{r=0}^{20} r^2 {}^{20}C_r = 20 \times 2^{18} (2 + 19) = 420 \times 2^{18}$$

$$\frac{d(UV)}{dx} = v \frac{dU}{dx} + U \frac{dv}{dx}$$

(Product Rule)

Result:

Productivity



QUESTION (JEE-MAINS 2020)

If the number of integral terms in the expansion of $(3^{\frac{1}{2}} + 5^{\frac{1}{8}})^n$ is 33, then the least value of n is

1 264 ~~X~~

2 128 ~~X~~

3 256 ✓

4 248 ~~X~~

(Sol) Here $T_{r+1} = {}^nC_r (3^{\frac{1}{2}})^{n-r} (5^{\frac{1}{8}})^r$

$\Rightarrow T_{r+1} = {}^nC_r 3^{\frac{n-r}{2}} (5)^{\frac{r}{8}}$

For integral terms, the set of possible of r is $\{0, 8, 8 \times 2, 8 \times 3, \dots, 8 \times 32\}$

Required value of $n = 8 \times 32 = 256$

QUESTION (JEE-MAINS 2026)

The sum of all possible values of $n \in \mathbb{N}$, so that the coefficients of x, x^2 and x^3 in the expansion of $(1+x)^2(1+x)^n$, are in Arithmetic Progression is :

Homework Problem.

1 12

2 7

3 3

4 9

QUESTION (JEE-MAINS 2026)

If $\left(\frac{1}{15C_0} + \frac{1}{15C_1}\right) \left(\frac{1}{15C_1} + \frac{1}{15C_2}\right) \left(\frac{1}{15C_2} + \frac{1}{15C_3}\right) \dots \left(\frac{1}{15C_{12}} + \frac{1}{15C_{13}}\right) = \frac{\alpha^3}{14C_0 \cdot 14C_1 \dots 14C_{12}}$, then the value of 30α is equal to _____

(Homework Problem)

A large, 3D rectangular box with a red face and yellow borders, tilted slightly to the right. It has a metallic, reflective appearance with highlights and shadows.

**MISCELLANEOUS
EXERCISE ON
CHAPTER – 7**

QUESTION

If a and b are distinct integers, prove that $a - b$ is a factor of $a^n - b^n$, whenever n is a positive integer. [Hint: write $a^n = (a - b + b)^n$ and expand]

Proof: Consider the following:

$$a^n = (a - b + b)^n = (t + b)^n, \text{ where } t = a - b.$$

Using Binomial Theorem;

$$a^n = {}^n C_0 t^n + {}^n C_1 t^{n-1} b + {}^n C_2 t^{n-2} b^2 + \dots + {}^n C_{n-1} t b^{n-1} + {}^n C_n b^n.$$

$$\Rightarrow a^n = (a - b)^n + {}^n C_1 (a - b)^{n-1} b + {}^n C_2 (a - b)^{n-2} b^2 + \dots + {}^n C_{n-1} (a - b) b^{n-1} + b^n$$

$$\Rightarrow a^n - b^n = (a - b)^n + {}^n C_1 (a - b)^{n-1} b + {}^n C_2 (a - b)^{n-2} b^2 + \dots + {}^n C_{n-1} (a - b) b^{n-1}$$

$$\Rightarrow a^n - b^n = (a - b) \left[(a - b)^{n-1} + {}^n C_1 (a - b)^{n-2} b + \dots + {}^n C_{n-1} b^{n-1} \right]$$

$$\Rightarrow a^n - b^n = (a - b) q$$

This shows that $a - b$ is a factor of $a^n - b^n$

Hence proved.

QUESTION



Evaluate $(\sqrt{3} + \sqrt{2})^6 - (\sqrt{3} - \sqrt{2})^6$.

H/W.

QUESTION

Find the value of $\left(\underline{a^2} + \sqrt{\underline{a^2 - 1}}\right)^4 + \left(\underline{a^2} - \sqrt{\underline{a^2 - 1}}\right)^4$.

HP/W

HP/W

$$(a-b)^4 + (a+b)^4$$

QUESTION



Find an approximation of $(0.99)^5$ using the first three terms of its expansion.

$$\text{H/w} \quad (1-0.1)^5 \approx T_1 + T_2 + T_3$$

QUESTION



Expand using Binomial Theorem $\left(1 + \frac{x}{2} - \frac{2}{x}\right)^4, x \neq 0.$

$$(a+b)^4.$$

QUESTION



Find the expansion of $(\underbrace{3x^2 - 2ax}_{a} + \underbrace{3a^2}_{b})^3$ using binomial theorem.

$$(a+b)^3.$$

H/w.

Thank
You